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RESEARCH ARTICLE

Soft Adaptive Feet for Legged Robots: An Open-Source Model for Locomotion Simulation

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ABSTRACT In recent years, artificial feet based on soft robotics and under-actuation principles emerged to improve mobility on challenging terrains. This paper presents the application of the MuJoCo physics engine to realize a digital twin of an adaptive soft foot developed for use with legged robots. We release the MuJoCo soft foot digital twin as open source to allow users and researchers to explore new approaches to locomotion. The work includes the system modeling techniques along with the kinematic and dynamic attributes involved. Validation is conducted through a rigorous comparison with bench tests on a physical prototype, replicating these experiments in simulation. Results are evaluated based on sole deformation and contact forces during foot-obstacle interaction. The model estimates the forces distribution on the foot sole with good accuracy, reaching a 10.5% error on the foot heel force estimation, 21.2% error on the foot modules, and 13.1% error on the foot metatarsus. The foot model is subsequently integrated into simulations of the humanoid robot COMAN+, replacing its original flat feet. Results show an improvement in the robot's ability to negotiate small obstacles without altering its control strategy. Ultimately, this study offers a comprehensive modeling approach for adaptive soft feet, supported by qualitative comparisons of bipedal locomotion with state of the art robotic feet.

INDEX TERMS Adaptive foot, digital twin, humanoid robots, legged robots, locomotion simulation.

I. INTRODUCTION

Replicating human locomotion and stability on uneven terrains has been a longstanding challenge in robotics. Flat-footed (e.g., ATLAS, Digit, HRP-4 [1], G1, Nao, Talos), and passive ball-shaped (e.g., ANYmal [2], HyQ [3], Spot) robots often struggle to traverse irregular terrains, limiting their applications in unstructured environments.

Humanoid robotic feet come in various structural designs, generally consisting of either a single rigid segment or two segments linked by a revolute joint. While effective on flat surfaces, these designs often lack the adaptability needed for uneven terrains. To address this limitation, more advanced designs have been introduced to enhance stability and agility. For instance, foot soles incorporating compliant visco-elastic materials [4], [5], [6] or airbag units [7], [8] can absorb small ground irregularities. Other designs use articulated structures made up of multiple segments joined by joints and compliant elements [9], [10], [11].

With simple foot designs, the challenge of locomotion on uneven terrain is primarily managed at the control architecture level. Common methods often rely on force/torque control [12] schemes and incorporate terrain recognition to actively adapt to the requirements of the environment. [13]. Other approaches utilize model predictive control [14] or integrate tactile sensors [15] to regulate foot-ground interactions. However, these strategies necessitate handling complex control mechanisms robustly. Our team took a different approach with the SoftFoot, an adaptive passive robotic foot. Designed for ground adaptability at a

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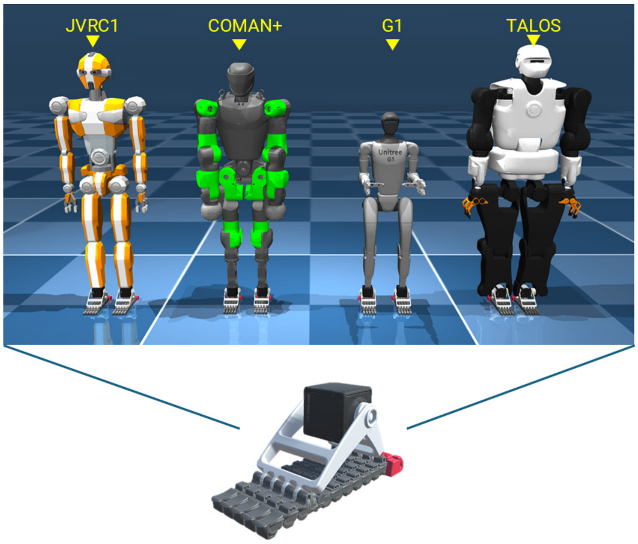


FIGURE 1. The SoftFoot digital twin (on the bottom) and its application to various legged robots in MuJoCo. From left to right: JVRC1, COMAN+, G1 from Unitree, and Talos from Pal Robotics.

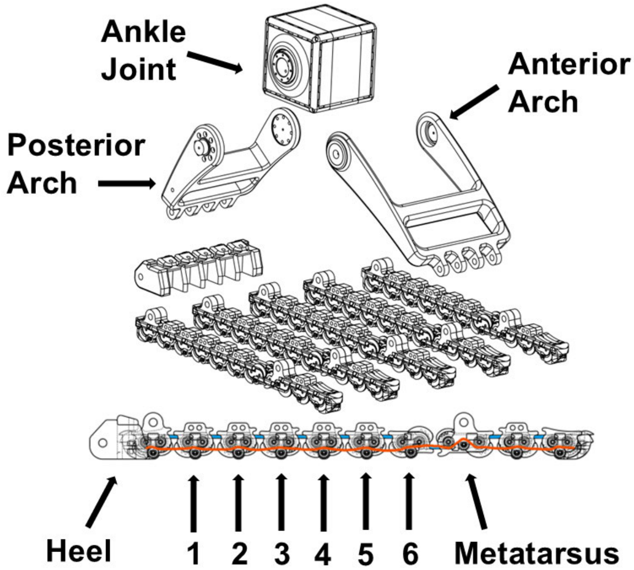


FIGURE 2. Exploded view of the robotic SoftFoot. The order of the plantar fascia modules is reported. The elastic bands in the sole are in blue and the tendon is in red.

mechanical level, it features a deformable sole that mimics the human foot [16]. Tests on the HRP-4 humanoid stepping on small obstacles demonstrated notable improvements in stability and obstacle negotiation [17]. Other studies took inspiration from the human foot to design rigid-flexible mechanisms, such as wheels, resulting in improved locomotion on rugged terrains [18].

Soft technologies offer enhanced adaptability, but their complexity poses substantial mathematical challenges. Analytical models often require simplifications to handle the dynamic interactions, uncertainties, and nonlinearities inherent in these systems. For example, ball-shaped feet are often reduced to a single ground contact point [19]. Physics engines and digital twins are powerful tools for simulating complex robotic systems [20], [21], providing dynamic virtual environments for design, real-time testing, and optimization [22], [23], [24]. MuJoCo, known for its speed and accuracy in constrained systems, is effective in simulating humanoid robots and their behavior on uneven terrains [25], [26]. Digital twins, as virtual replicas of physical systems, enable real-time simulation, performance prediction, and iterative refinement. For the SoftFoot, a digital twin allows accurate testing of its adaptability and stability on uneven terrain without extensive physical prototyping.

In this work, we present a digital twin of the robotic SoftFoot, developed within the MuJoCo physics engine (Fig. 1), and its application in legged robot locomotion. This model enables the analysis of SoftFoot's interaction with uneven terrain and obstacles while seamlessly integrating with MuJoCo-based legged robot models (see Fig. 1). By leveraging simulations, we can investigate the potential of soft adaptive feet in locomotion. The model supports studies on foot adaptability based on geometry, the relationship between applied force and foot conformation, active step planning, the integration of locomotion adaptiveness at both software and hardware levels, and the exploration of reinforcement learning strategies. To support further research, we release the SoftFoot model as open-source within the Natural Machine Motion Initiative (https://github.com/NMMI/SoftFoot_MuJoCo) and on the freely available MuJoCo model collection (https://github.com/google-deepmind/mujoco_menagerie) [27], allowing the community to develop and test new locomotion approaches.

This paper details the model description (Section II) and validation (Section III) by comparing simulations with bench tests on the physical prototype [28]. Accuracy and performance are assessed through sole deformation and contact forces. We then simulate a humanoid walking with SoftFoot and its original feet (Section IV) [17] to analyze obstacle negotiation. Finally, we discuss the model's accuracy, effectiveness, and limitations (Section V).

II. SOFTFOOT MODEL

A. THE SOFTFOOT

The SoftFoot is composed of simple, compact, and modular mechanical structures [16]. The SoftFoot features a posterior arch rigidly connected to the ankle joint, and an anterior arch connected to the posterior with a revolute joint (Fig. 2). Five modular and flexible parallel structures called *chains*, form the deformable sole, which acts as the *plantar fascia* (a thick band of tissue running across the bottom of the foot from the medial process of the calcaneal tuberosity to the bases of the proximal phalanges) and *toes* of the human foot. Each of these structures is designed with a series of modular components, which will be referred to as *module* from here onwards,

TABLE 1. Components of the SoftFoot and their corresponding materials, indicating whether each was manufactured or commercially sourced.

	Material / Product	Manufactured / Commercial
Cube	qbmove Advanced	Commercial (qb Robotics)
Frontal Arch	ABS	3D printed
Posterior Arch	ABS	3D printed
Modules	Nylon	Injection Molding
Heel	ABS	3D printed
Tendons	LIROS DC 0.6 mm	Commercial (LIROS GmbH)
Elastic Bands	Natural rubber (35 ShA)	Commercial

similar to those developed for the Pisa/IIT SoftHand [27], engineered to roll on each other utilizing a specific flexible configuration of the Rolamite joint. The modules are secured together with two elastic bands for each coupling. A tendon runs through each of the five chains, following a specific route as depicted in Fig. 2, to distribute forces between joints. The *metatarsus* module is specifically designed to allow the connection of the plantar fascia with the anterior arch through a revolute joint.

The SoftFoot sole adapts its shape and stiffness through a system of pulleys, tendons, and springs in response to the applied load and ground profile. It remains compliant under low or no load but stiffens when a significant load is applied due to tendon stretching, similar to the human foot's plantar fascia. This architecture enables adaptability, impact absorption, and stabilization.

A complete list of the components and materials of the physical prototype is provided in Table 1, whereas the physical properties, such as mass and inertia, are specified directly within the model.

B. THE MUJOCO SOFTFOOT MODEL

MuJoCo allows us to model all the SoftFoot components effectively. We extracted size, position, and inertia data from the CAD model and exported simplified meshes. The foot structure is defined as two open chains (Fig. 3(b)): one from the ankle joint, including the posterior arch, heel, and five sole chains; the other comprising the anterior arch. These chains connect at the metatarsal joint (red circle in Fig. 3(b)) via an equality constraint, forming a closed-chain structure. Finally, we modeled the foot sole components and their functioning:

1) MODULES

The SoftFoot modules present mechanical limit switches to limit the rotation between couplings. This effect is modeled by setting a revolute joint range of 20–90 degrees for anticlockwise and clockwise rotation, considering the foot oriented as in Fig. 2. The module's coupling with the Rolamite joint is modeled with the assumption of pure rotation without sliding or losing contact. This is obtained by imposing an equality constraint on the relative rotation between adjacent modules in a coupling (see angle β Fig. 3(c)). This assumption holds for the prototype, except under high loads, where modules may lose contact. Such behavior can be replicated in MuJoCo using soft constraints and fine-tuning equality constraint parameters.

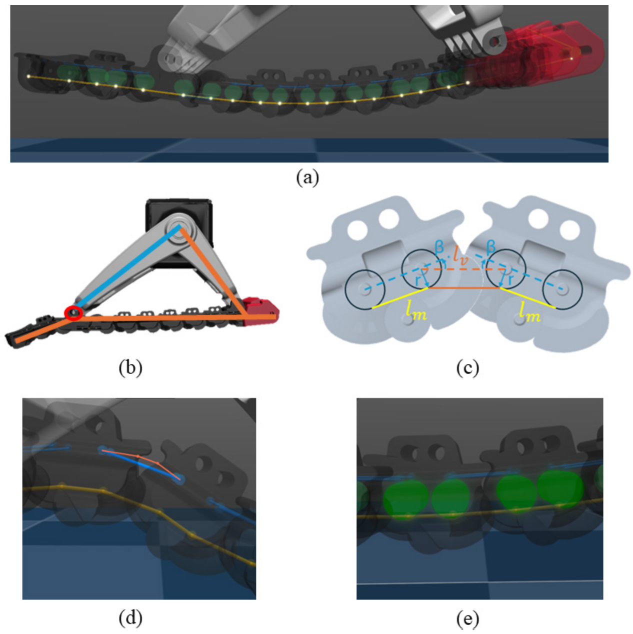


FIGURE 3. MuJoCo modelling techniques for the SoftFoot. (a) The tendon in the foot sole (in yellow) passes through specific points (in white), defined for each module to reproduce its routing. (b) Two open-chain structures (blue and orange) form the foot model, with the red circle marking their connection. (c) Rotation equality constraint between coupled modules and the resulting tendon length change: when one module rotates by an angle β the coupled module also rotates by an angle β in the opposite direction. (d) Elastic band paths in simulation (blue) vs. real prototype (orange). (e) Close-up of virtual pulleys added to each module.

2) ELASTIC BANDS

Two distinct approaches are possible to model the elastic bands connecting coupled modules: using MuJoCo's elastic elements or incorporating their properties into joint parameters. MuJoCo's elastic tendons effectively replicate band behavior by acting as unidirectional tension springs, anchored to coupled modules. As the coupling rotates, the elastic element extends and applies a force to bring the modules back to their resting position. The resting length and extension range come from band datasheets, while stiffness is estimated using geometric properties and the Gent equation [29] from Shore A hardness. Alternatively, the elastic band characteristics can be transferred to the coupling joint parameters without modeling additional elastic elements. The joint stiffness is obtained by transposing elastic band stiffness to the corresponding joint angular stiffness.

However, the first modeling approach introduces a deviation from the real prototype. Since the linear elastic elements lack direct contact with the module, their path differs from the actual elastic bands, which follow predefined internal routes affecting their stretch (Fig. 3(d)). Additionally, modeling all elastic elements requires 45 elements and 90 extremities, increasing its complexity. Therefore, we modeled the elastic properties at the coupling joints.

3) TENDONS

We added virtual massless cylindrical geometries in the modules to act as pulleys to accurately replicate the tendon routing and its force distribution functionality (see Fig. 3(e)). The tendon in MuJoCo is defined by the minimal path between designated points. We set these points in the reference frame of the pulleys so that they rotate accordingly to the module, consequently modifying the contact point between the tendon and the pulley. Therefore, the path and the lengths of the tendons are determined by the shape of the ground, i.e., the rotation of the sole modules. Referring to Fig. 3(c), the length of the tendon:

$$L = n * l_m + \sum_{i}^{n+1} (l_v - 2r \sin \beta_i) \tag{1}$$

where L is the length of the tendon, n the number of modules, l_m is the length of the tendon part connecting the two pulleys of the same module, l_v is the length of the tendon part connecting the pulleys of two adjacent modules when the modules have no rotation, r is the radius of the pulleys, and β_i is the rotation of the i th module. The prototype's tendons are almost inextensible, and modeled by imposing the admissible tendon length range in the model. The resting length of the tendon element was heuristically obtained from the simulation. Fig. 3(a) illustrates one of the tendons in the model and its route along the sole.

MuJoCo employs a soft contact model, where simulated bodies interpenetrate and generate forces based on contact depth. In our model, elastic bands keep the module bodies in contact, generating interpenetration forces that destabilize the simulation. We excluded inter-module contacts to prevent this, as they do not affect the foot's functionality. The resultant model is shown in the video attachment.

III. MODEL VALIDATION

To verify the accuracy of the SoftFoot model, we reproduced in MuJoCo the experiments performed in [28] to validate an analytical model of the same SoftFoot prototype.

A. EXPERIMENTAL SETUP

In the experiments, a vertical load was applied to a SoftFoot with a single-chain sole in contact with three sensorized obstacles. Two obstacles were placed under the heel and metatarsus, while a third, height-adjustable obstacle aligned with the center of each of the six plantar fascia modules (Fig. 2) across different trials. Each module was tested with obstacle heights of 7 mm, 11 mm, 15 mm, and 19 mm. Contact forces were measured using six-axis force/torque sensors, and foot rotations were tracked with IMUs attached to each sole module.

This setup was replicated in MuJoCo (Fig. 4). The foot was mounted on a vertically actuated slider, initially positioned to avoid contact. As the simulation ran, gravity and the applied load pressed the foot down until equilibrium was reached. In this virtual setup, contact forces were recorded using MuJoCo sensors, and module rotations were extracted from simulation data. The virtual setup is shown in the video attachment. The simulations were performed using the default MuJoCo timestep of 0.002 s, as using a smaller timestep did not show accuracy improvements.

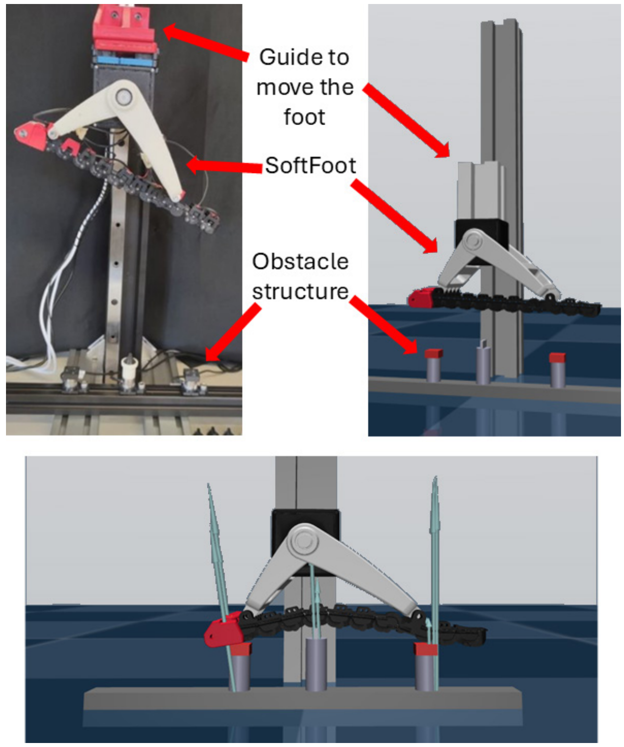


FIGURE 4. Experimental setup. On the left, the setup from the experiments in [28], on the right, the simulated setup in MuJoCo. At the bottom, a representation of the tests performed with the blue arrows representing the forces.

B. METHODS

Each trial lasted 5 seconds, enough to reach equilibrium given the initial foot-obstacle distance. Forces and rotations were recorded from the final simulation step. A total of 48 trials were conducted, testing two loads (approximately 12 N and 24 N) across four obstacle heights and six positions.

The forces obtained from the experimental sessions were compared with those calculated from the simulation (represented in Fig. 4). Only the vertical component of the force was considered, as it represents the major component of the total force, while the other force components were not taken into account, leaving this analysis for more specific future investigations. Relative errors on the forces were computed for each of the three foot components in contact with an obstacle: the heel (h), the metatarsus (m), and the interested plantar fascia module (p). In particular, for each body $b \in \{h, p, m\}$ the relative error on the vertical force was calculated as:

$$e_b = (\hat{f}_b - \tilde{f}_b)/\tilde{f}_b$$

where $\hat{f}_b$ and $\tilde{f}_b$ are respectively the force measured in the simulation and recorded in the experimental session. The error on the total vertical force acting at the three body $b \in$

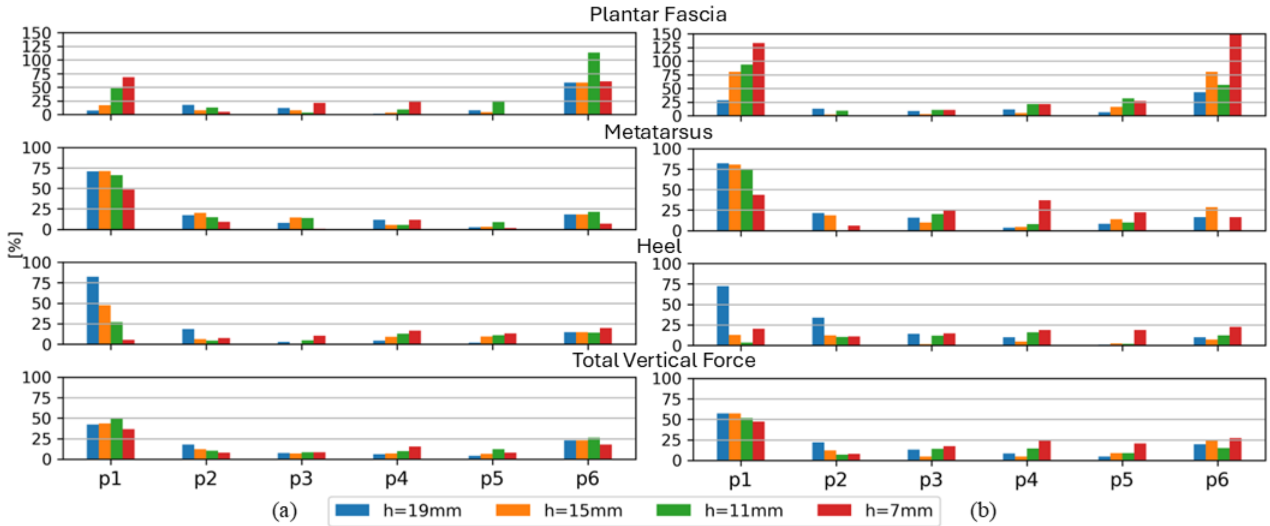


FIGURE 5. Force relative errors for the three foot parts in contact with the obstacles and the error on the total vertical force. Positions (p) 1 and 6 represent the position closer to the heel and to the metatarsus module, respectively. Results are divided according to the applied load, (a) 12 N **and (b)** 24 N **respectively.**

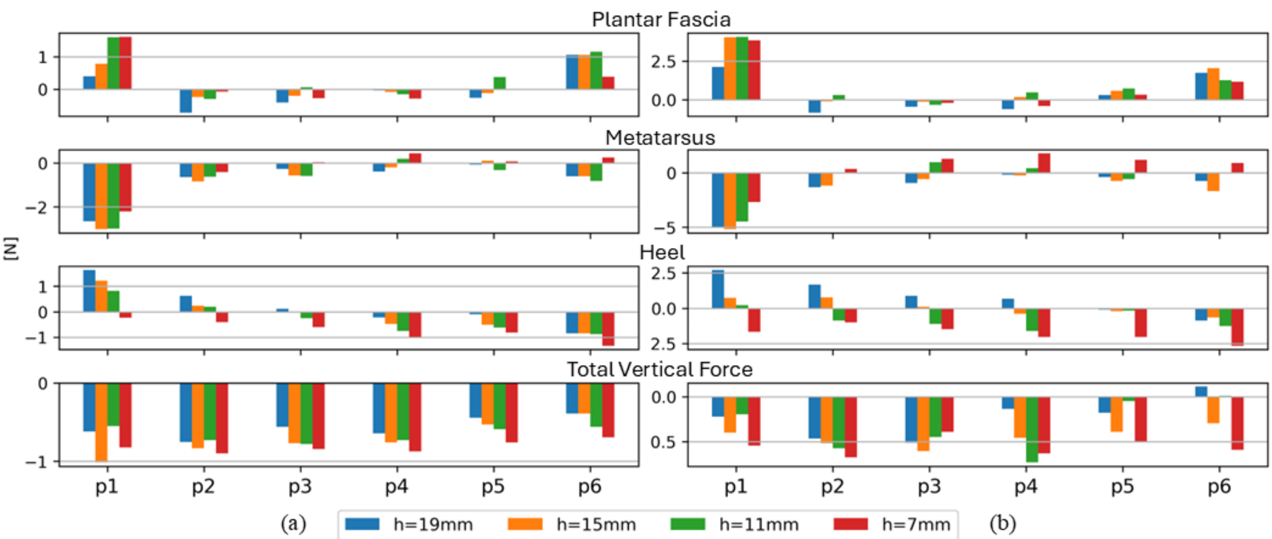


FIGURE 6. Force errors for the three foot part in contact with the obstacles and the error on the total vertical force. Positions (p) 1 and 6 represent the position closer to the heel and to the metatarsus module, respectively. Results are divided according to the applied load, (a) 12 N **and (b)** 24 N **respectively.**

$\{h, p, m\}$ was calculated as:

$$e_T = \sum (\hat{f}_b - \tilde{f}_b) / \sum \tilde{f}_b$$

Module rotations from the simulation were compared to experimental data, focusing on the sagittal plane, where SoftFoot has enhanced ground adaptability. Differences in IMU placement and body reference frames introduced an offset, which was removed for accurate comparison. This offset was calculated as the mean difference between simulated and experimental rotations.

The calculated error for the forces acting on the modules and rotations defines the model's accuracy, i.e., how closely the model's force distribution and dynamics agree with those measured with the real device in the same scenario.

C. RESULTS

1) FORCE VALIDATION

Fig. 5 presents the absolute values of the relative errors in the forces measured on the three obstacles under each experimental condition for the two applied loads, and defines how close the force measured in the simulation is to the one measured in the real scenario in terms of percentage. Conversely, Fig. 6 illustrates the absolute errors, defined as the difference between the forces in the simulation and those recorded in the experiments considering their value in Newton. A positive value indicates that the force in the simulation is higher than the experimental, while a negative value implies that it is lower.

2) SOLE DEFORMATION VALIDATION

Fig. 7 displays the discrepancies in the rotation of the different plantar fascia modules in the sagittal plane between the experimental data and the simulations, with the results divided according to the applied load, thus describing how close the dynamic behavior of the foot sole in the model is compared to the sole of the real device. Fig. 8 shows the statistical distribution of rotation errors for each module across all simulations, summarizing the results from Fig. 7.

IV. BIPEDAL ROBOT LOCOMOTION

A. PRELIMINARY BENCH TEST

Before running the locomotion simulation with a legged robot, we tested the model on the same bench test setup used for its validation with a load comparable to that of a humanoid robot. To assess the functionality of the foot with higher loads, comparable to those of a humanoid robot, we run simulations using the same setup and obstacle conditions used for the model validation, and assuming an even distribution of the forces on the five foot sole chains, we applied a 15 kg load with the slider (considering a weight of 75 kg for the robot). As the experiments carried out in [28] do not consider high loads, we compared the simulation forces to those of the simulation performed with a 12 N load to get an overview of the force distribution. The force data from simulations are normalized on the applied load to make the two conditions comparable. Fig. 9 shows the normalized force for each tested obstacle condition for the heel, metatarsus and plantar fascia element of the foot. The force distribution on the three elements presents a similar trend with both loads, although the force is higher on the metatarsus element with the higher load. This is probably caused by the stress-stiffening behavior of the tendon, which distributes forces more evenly on the sole as the load increases.

B. EXPERIMENTAL SETUP

To evaluate qualitatively the functionality of the SoftFoot model, we simulated the torque-controlled humanoid COMAN+ [30] walking over 5 mm and 10 mm square-section obstacles (Fig. 10). Tests used both the original rigid feet and the SoftFoot without altering the control structure, following prior work with HRP-4 [17]. Obstacles were randomly placed in all trials. Simulations ran on an Intel i7-12700H CPU with an NVIDIA GeForce 3060 Laptop GPU. We added box geometries encapsulating the modules as colliders to simplify ground collisions and improve simulation stability. Nevertheless, given the high number of contacts with the ground due to the articulated structure of the SoftFoot and the constraints used to model its behavior, it was necessary to decrease the simulation timestep to 0.1 ms when using the SoftFoot. Higher timesteps resulted in an unstable simulation.

C. METHODS

The robot walks with a whole-body non-linear Model Predictive Control (MPC) using the Horizon framework [31] that generates position, velocities, and torque references, minimizing a cost function that considers Cartesian references. By setting the Cartesian references and the gait sequence and timing, the MPC directly outputs the joint space references that are interpolated to match the low-level control frequency (i.e., 1 kHz).

Unlike the HRP-4 experiments, the SoftFoot was not designed for COMAN+; we simply replaced its original feet. The SoftFoot differs significantly in size, measuring 27 cm long (16 cm within the arches) and 9 cm wide, compared to the rigid foot's 21 cm length and 11 cm width. It is also asymmetrical in the frontal plane. To accommodate these differences, the robot's ankle position was adjusted 1 cm posteriorly and 2 cm downward to ensure stability. A detailed simulation view is available in the video attachment.

D. RESULTS

The simulations show that the humanoid robot can overcome small obstacles without requiring a specific control structure when using the SoftFoot. In contrast, it tends to fall when using its original rigid flat foot. This difference is further depicted in Fig. 11, which shows how the two types of feet interact with the obstacle. While the rigid foot generates a very small contact surface with the obstacle and the ground, resulting in a small support area and so a poor region of stability, the SoftFoot filters the obstacle, optimizing the contact surface and the support area, by being in contact with the obstacle with the plantar fascia region and the ground with the heel and the toes. Moreover, by changing the number and place of contact points, the force distribution on the foot sole also changes.

V. DISCUSSION

A. MODEL VALIDATION

1) FORCE VALIDATION

The absolute error in total vertical force remains below 10% of the applied load across conditions, primarily due to a lower force in simulations. This discrepancy arises because we used the nominal force declared in the experiment, while the experimental setup involved manual weight placement, introducing variability.

Force errors follow a consistent pattern across foot components, with higher errors when the moving obstacle is near the heel (p1) for all three sole parts. As shown in Fig. 6, these errors appear to stem from the distribution of the load on the sole. Specifically, an excessive force (determined by the positive value of the error) is generated by the module of the fascia and the heel, consequently resulting in less force from the metatarsus (negative error). Errors decrease at other positions, staying below 24.2% for the lower load and 31.4% for the higher, except at p6, where the fascia module exerts excessive simulated force. While absolute error is higher at

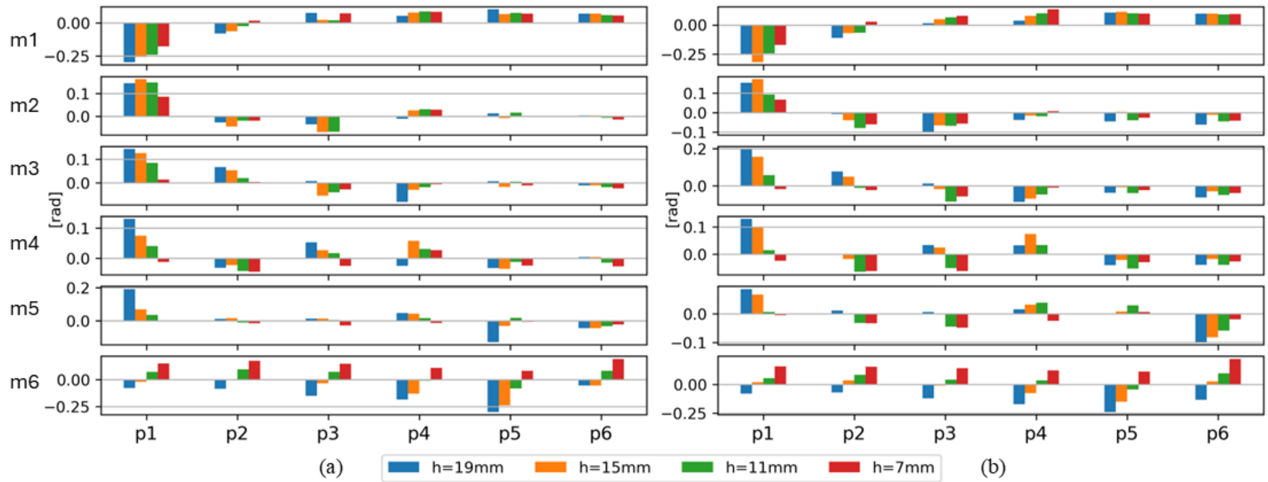


FIGURE 7. Rotation errors for the different modules composing the plantar fascia (m1-6). Positions (p) 1 and 6 represent the position closer to the heel and to the metatarsus module, respectively. Results are divided according to the applied load, (a) 12 N and (b) 24 N respectively.

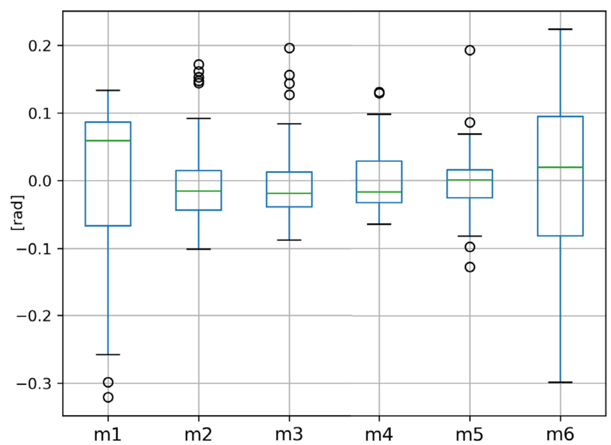


FIGURE 8. Box plot representation of the rotation error for each of the six plantar fascia modules (m1-6) for all the 48 trials. The green line represents the median, and the box is the interval between the first and the third quartile. The black circles correspond to the outliers.

p1, relative error at p6 increases due to the lower experimental force magnitude.

Across obstacle heights, relative error trends higher for plantar fascia modules at the lowest height (7 mm) (Fig. 5, Fig. 6), as lower forces amplify relative error. We argue that the lack of sensitivity to low forces does not represent an issue for robotic applications like humanoid locomotion, where foot loads are considerably higher. The measured force might depend on where the obstacle contacts the fascia modules. However, the setup was designed to touch the middle of each module, and we replicated the setup following this principle in the simulations.

Table 2 summarizes the results obtained in terms of mean values of the force relative error for each component, considering all 48 simulations. The mean relative force error is 14.6% for the heel, 33.2% for the plantar fascia modules, and 21.7% for the metatarsus.

TABLE 2. Mean relative errors of contact forces estimation with the MuJoCo and the analytical [28] models.

	HEEL	MODULES	METATARSUS	TOTAL FORCE
MuJoCo	14, 6%	33, 2%	21, 7%	18, 6%
MuJoCo Filtered	10, 5%	21, 2%	13, 1%	11, 8%
Analytical	31, 0%	59, 5%	7, 2%	47, 3%
Analytical Filtered	22, 4%	54, 5%	5, 2%	46, 8%

When reconstructing the experimental setup, uncertainties stemmed from heel and metatarsus obstacle positions and initial SoftFoot orientation. Small changes (±4mm) in heel and metatarsus obstacle placement altered mean force errors by up to 13%. Adjusting initial inclination (1∘–5∘) changed errors by 14% for the heel, 19% for the plantar fascia, and 20% for the metatarsus. To mitigate these uncertainties, we applied a mean + standard deviation filter to the relative force error to remove possible outliers, reducing the mean relative force error to 10.5% for the heel, 21.2% for the plantar fascia, and 13.1% for the metatarsus (see Table 2).

Compared to the analytical model developed by Mura et al. [28], the MuJoCo model improved the force evaluation accuracy. The errors in our simulations are generally lower for the plantar fascia modules. While errors near the metatarsus are higher, heel and metatarsus obstacle errors are comparable. Quantitatively, the mean relative error in the analytical model is 31.0% for the heel, 59.5% for the plantar fascia modules, and 7.2% for the metatarsus. Applying the same type of filter we used for our results, these errors are 22.4%, 54.5%, and 5.2%, respectively. The force estimation errors are reported in Table 2. Although the error in the model estimation of contact forces does not directly translate into the foot performance, it is essential to define how accurately the model reproduces the force distribution on the real foot in the same scenario. Table 3 summarizes the modeling strategy options and simulation trade-offs, indicating whether they affect the simulation force distribution accuracy.

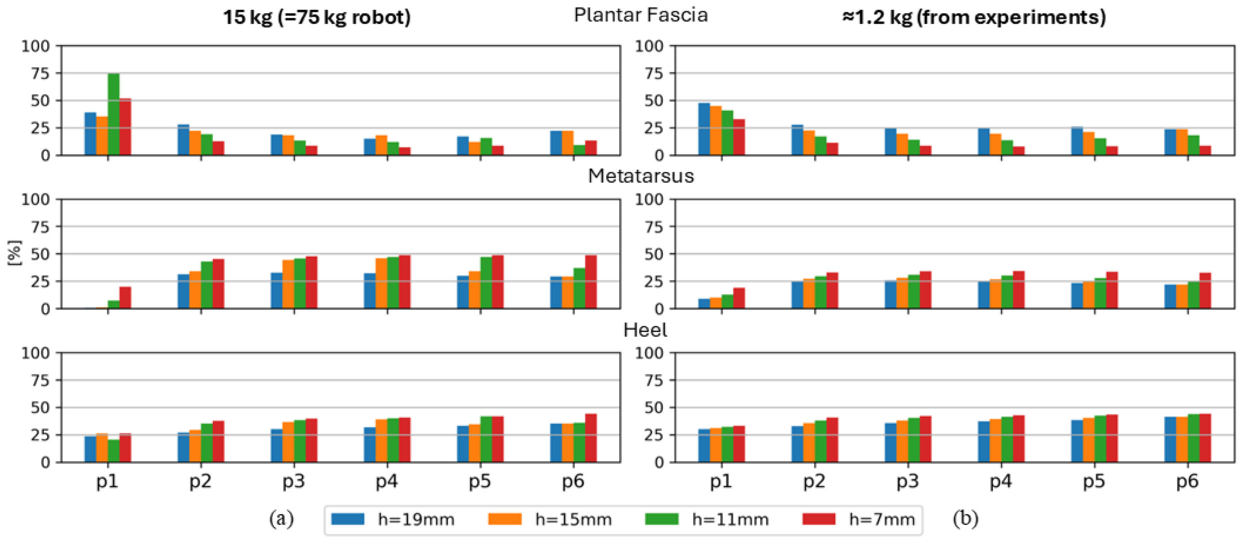


FIGURE 9. Normalized force on the plantar fascia module, metatarsus and heel across the different obstacle positions and heights for a 15 kg load (a 75 kg robot) (a) and 1.2 kg load (b) on a single chain.

TABLE 3. Summary of the modeling strategies and simulation trade-offs, indicating if they affect the simulation force accuracy.

	Modeling/Trade-offs		Affects Accuracy
Elastic bands	Physical tendon	Joint properties	X
Modules box colliders	No	Yes	X
Time step bench test	Default (0.002 s)	Lower	X
Time step bipedal	Default (0.002 s)	0.0001 s	V

2) SOLE DEFORMATION VALIDATION

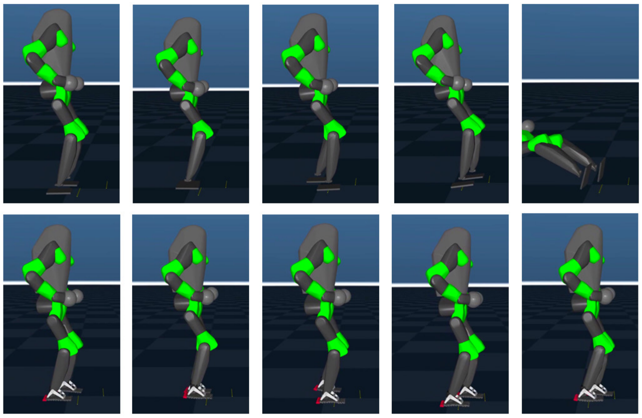
As shown in Fig. 7, rotational error for plantar fascia modules is generally small, though it varies across modules. Similar to force errors, rotation error increases when the obstacle is closer to the heel. Fig. 8 summarizes the results across all conditions. Higher errors in m1 and m6 may stem from the pure rotation assumption between coupled modules and the difficulty in placing the IMUs for those modules in the experiments. While rotation error fluctuates across trials, it remains below 0.3 rad, with most values between −0.1 rad and 0.1 rad (Fig. 8). For m2 to m5, the full error range stays within this limit. Due to the low loads used in this experiment, stress-stiffening behavior is not visible in Fig. 5 and Fig. 7. However, it is an intrinsic characteristic of the foot, and it can be easily shown by simulating the foot under a considerable vertical load.

B. LOCOMOTION ASSESSMENT

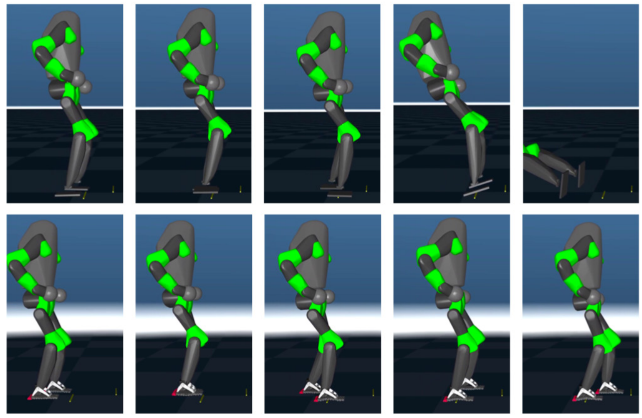
Fig. 10 illustrates COMAN+ walking over obstacles using both its original rigid foot and the SoftFoot. The rigid foot acts as a cantilever, and given the height difference in the contact point with the ground and the obstacle, the robot's center of mass is projected outside the support area between the feet. The higher the obstacle, the more the motion of the center of mass increases, and the more the foot support region diminishes (see [16] for the mathematical analysis of how the foot sole characteristics affect the stability). The SoftFoot ability to filter the obstacle significantly reduces this phenomenon (see [28] for its experimental demonstration).

Although the robot was sometimes able to maintain balance with the rigid feet on 5 mm obstacles, it consistently fell on 10 mm obstacles. In contrast, the SoftFoot generally handled both obstacle heights without issue, though the outcome depended on where the sole touched the obstacle. The obstacle dimensions and shapes match those from the experimental session for validation. This confirms the model's ability to adapt to tested obstacles. Based on its geometry, the foot can handle obstacles up to 70 mm high in the sole position aligned with the ankle joint while maintaining heel and toe contact. However, the generalization of SoftFoot's adaptability is yet to be defined, and the model serves as a valuable tool for this investigation. The most robust performance was observed when the contact occurred between the posterior and anterior arches, which is the most flexible part of the sole. Designing motion planners that take advantage of the properties of the SoftFoot may also lead to more stable locomotion. If an unavoidable obstacle is identified, then the robot's footfall could be adjusted so that it can land in the area of the sole with the most compliance. Therefore, studying the joint benefits of embodied robustness through compliant mechanisms and adaptive control is a future research goal. In particular, compliant end effectors can allow less effective controllers to succeed as well. Reinforcement learning could further accelerate policy optimization, enabling faster convergence in both simulation and real-world applications.

The same adaptive foot model was also used with a human-like model, controlled via a reinforcement learning policy based on human walking data, to test the differences and advantages between rigid and adaptive feet [32]. Tests comparing a rigid and the adaptive foot across various terrains, slopes, and obstacles demonstrated the benefits of



(a) 5 mm



(b) 10 mm

FIGURE 10. Photo-sequence example of the COMAN+ MuJoCo model (represented in collision geometries) walking on an obstacle, 5 mm in (a), 10 mm in (b), with its own feet (top) and the SoftFoot (bottom). Notice how the robot can negotiate obstacles when equipped with SoftFoot, while losing balance when using its rigid flat feet.

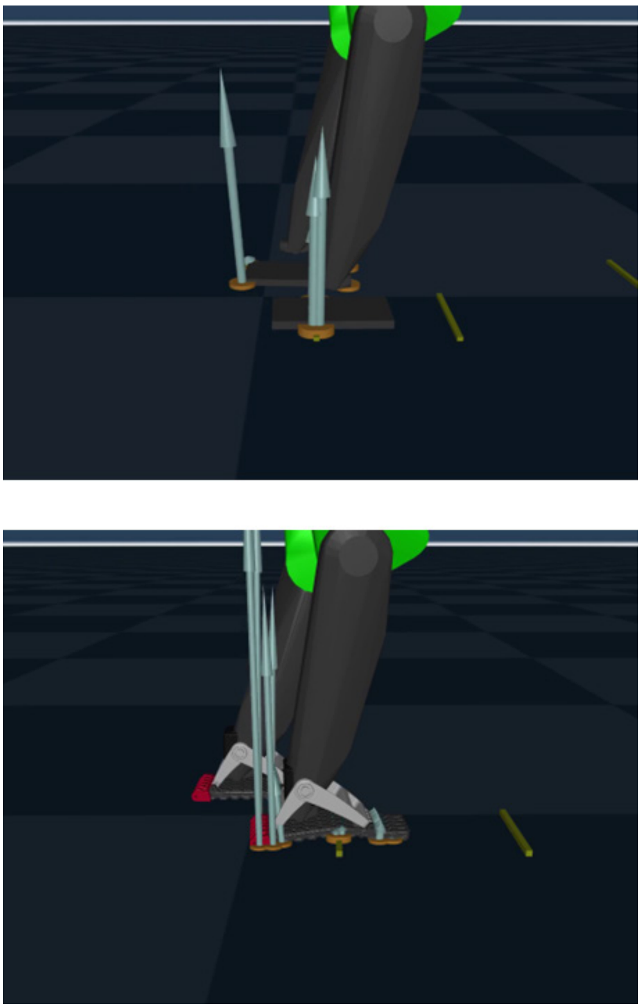


FIGURE 11. Close-up view of COMAN+ addressing the obstacle with its rigid foot on the top and the SoftFoot on the bottom. Orange points represent the foot contact points, while the forces are represented by the light blue arrows with a dimension proportional to their module.

a compliant sole. Simulation data were then compared to the real-life scenario.

Integrating the SoftFoot digital twin with the COMAN+ model was straightforward, requiring only a substitution in the model's code for the foot description. This integration was similarly seamless with other bipedal robots, as shown in Fig. 1. However, since the foot is not specifically designed for COMAN+ or any particular robot, minor adjustments in the attachment may be needed. Moreover, due to its biomimetic design, the foot model can be readily applied to prosthetics and rehabilitation robotics research. For instance, in [32], the model was integrated into a human model to replicate human-like gait. Potential applications of this simulation framework range from the design of novel prosthetic devices to the development of advanced biomimetic control strategies.

C. LIMITATIONS

This work presents some limitations, which are summarized below. First, the bench-test validation of the foot was conducted under low-load conditions, which do not reflect the actual weight of the robot. Although simulations with higher loads were performed, it was not possible to validate them against real-world data. Consequently, the accuracy of force distribution estimation and foot module rotation accuracy may differ under higher loads. Second, the locomotion assessment was limited to a qualitative evaluation. While it demonstrated the basic functionality of the foot, it did not allow for quantitative comparisons with real-world data to establish accuracy. A more detailed investigation of model scalability under higher load conditions (beyond the simulated 75 kg) would enhance applicability to real humanoids. Third, the analysis of computational cost was restricted to identifying the minimum timestep required for stable simulation. A notable slowdown in locomotion simulation was observed, but its underlying causes remain unclear. This issue was not reported in other works, such as [32], which suggests that the slowdown may be specific to the present simulation environment. Possible contributing factors include the high number of contacts, the complexity of the foot model, the concurrent execution of the real-time control framework, or a combination of these. A more in-depth analysis is required to confirm this hypothesis.

VI. CONCLUSION

This study introduced a validated digital twin of the SoftFoot, developed with MuJoCo and released as an open-source tool for exploring new locomotion approaches. The model accurately captures key characteristics of the SoftFoot, including its ground compliance and the ability to stiffen under load, and provides reliable contact force estimates. However, the model accuracy was validated with lower loads compared to the weight of a bipedal robot, and the force estimation might be affected under higher loads, although in this work, we proposed a simulated comparison of the force distributions under 75 kg to verify the functionality of the model. We demonstrated its integration with MuJoCo legged robots, highlighting its potential to improve stability and adaptability in humanoid locomotion. However, simulation complexity slowed performance, and the model required attachment adjustments to align the center of mass with the ankle joint.

Future work will focus on simplifying the model to improve simulation speed while maintaining accuracy, potentially integrating it with the analytical model from [28] for better force estimation. The model will be evaluated across various scenarios to assess its computational cost, simulation speed, and applicability limits. Its accuracy will be validated under higher loads comparable to those of a real bipedal robot. Furthermore, the model will support an in-depth analysis of real-time implementation, incorporating additional quantitative performance metrics, such as stability margins and gait efficiency, to complement qualitative results. We also plan to explore new control strategies, such as footstep planning that directs contact toward the plantar fascia area inside the foot arches, avoiding the heel and metatarsus. Additionally, we aim to expand locomotion simulations to diverse environments and conditions to further assess SoftFoot's adaptability. This digital twin establishes a foundation for future foot designs, facilitating innovations that could significantly advance the capabilities of humanoid robots in real-world applications.

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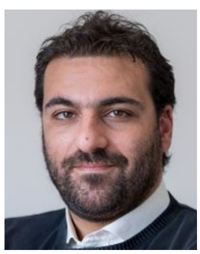

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